WORKED SOLUTION

STEP III 1988, Question 5

Pure • Recurrence Relations and Loans

STEP III 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q5

The key idea

Write down what happens to the debt in one day. Each day the debt is multiplied by $1+k$ (interest), and then a fixed amount is either added (digging cost) or taken away (earnings). A recurrence of the form $S_n = (1+k)S_{n-1} +$ constant unrolls into a geometric series. The final part then comes down to one comparison: is the daily income bigger than the interest on the debt at the moment the tunnel opens?

Solution

The debt while the tunnel is being dug

An interest rate of $100k\%$ means the daily interest on a debt of £S is £kS. Let S_n be the debt (in pounds) at the end of day n, with $S_0 = 0$. At the end of day $n \le T$ the firm borrows c for the digging and kS_{n-1} for the interest, so

$$S_n = S_{n-1} + kS_{n-1} + c = (1+k)S_{n-1} + c.$$

Applying this repeatedly, starting from $S_0 = 0$,

$$\begin{aligned} S_n &= c + (1+k)S_{n-1} \\ &= c + c(1+k) + (1+k)^2 S_{n-2} \\ &= \cdots = c[1 + (1+k) + (1+k)^2 + \cdots + (1+k)^{n-1}]. \end{aligned}$$

This is a geometric series with n terms and common ratio $1+k \neq 1$ (since $k > 0$), so

$$S_n = \frac{c[(1+k)^n - 1]}{k}, \qquad n \le T,$$

as required. (To be careful, check it by induction: it gives $S_0 = 0$, and if it holds for S_{n-1} then $(1+k)S_{n-1} + c = \frac{c}{k}[(1+k)^n - (1+k) + k]$, which is the formula for S_n.)

The debt after the tunnel opens

From day $T+1$ on there is no more digging. Each day the debt gains interest kS and the firm pays in d, so, for as long as the debt is still positive,

$$S_{T+j} = (1+k)S_{T+j-1} - d.$$

This is the same recurrence as before with c replaced by $-d$, but now it starts from S_T instead of from 0. Unrolling it m times,

$$\begin{aligned} S_{T+m} &= (1+k)^m S_T - d[1 + (1+k) + \cdots + (1+k)^{m-1}] \\ &= (1+k)^m S_T - \frac{d[(1+k)^m - 1]}{k}. \end{aligned}$$

Substituting S_T from the first part,

$$S_{T+m} = \frac{c[(1+k)^T - 1](1+k)^m - d[(1+k)^m - 1]}{k}.$$

We need $S_{T+m} > 0$ because the recurrence only applies while the debt is still being paid. On the day the debt is cleared, the firm pays only what is owed, not the full d.

When the debt is paid off

Collect the terms in $(1+k)^m$:

$$S_{T+m} = \frac{d}{k} - \frac{(1+k)^m}{k}\Big(d - c[(1+k)^T - 1]\Big).$$

Ask yourself What does the condition $d/c > (1+k)^T - 1$ actually mean in money terms?

Multiply it by c: it says $d > c[(1+k)^T - 1] = kS_T$. In other words, the first day's income is more than the interest on the debt when the tunnel opens. So the debt goes down on day $T+1$, and the interest then goes down too, so it keeps going down faster and faster. The algebra makes this precise.

Suppose $d > c[(1+k)^T - 1]$, and write $\delta = d - c[(1+k)^T - 1] > 0$. Then

$$S_{T+m} = \frac{d - \delta(1+k)^m}{k}.$$

Since $1+k > 1$, $(1+k)^m \to \infty$ as $m \to \infty$, so $\delta(1+k)^m$ eventually exceeds d. The formula is valid only while the debt is positive, so it can't stay valid for ever: there is a first day on which the remaining debt is at most d plus that day's interest, and on that day the firm pays it off. In fact the debt is cleared on day $T+m$ for the least m with

$$(1+k)^m \geq \frac{d}{\delta}, \qquad \text{that is,} \qquad m \geq \frac{\ln(d/\delta)}{\ln(1+k)}.$$

So, if $d/c > (1+k)^T - 1$, the firm eventually pays off the debt, as required.

The condition is also necessary. If $d \le kS_T$, then $\delta \le 0$, and since $S_T = (d-\delta)/k$,

$$S_{T+m} - S_T = \frac{-\delta[(1+k)^m - 1]}{k} \ge 0$$

for every m. The debt never falls below S_T, so it is never paid off.

Checks

- Put $n = 1$: $S_1 = c$, which is right since nothing was owed on day 1, so no interest is due.
- Put $m = 0$ in the formula for S_{T+m}: it gives S_T, as it should.
- If $d = kS_T$ exactly, then $\delta = 0$ and $S_{T+m} = d/k = S_T$ for every m: the income just covers the interest and the debt never moves. This is the boundary case.
- A day-by-day simulation with $c = 100$, $k = 0.01$, $T = 30$ gives $S_T \approx 3478.49$, and the formula agrees. The threshold is $d > 34.78$. With $d = 50$ the debt is cleared after $m = 120$ days, matching $\ln(50/15.22) / \ln 1.01 \approx 119.6$.

Common slips

- Charging interest on day 1, or counting $n+1$ terms in the series. With $S_0 = 0$ the first day costs exactly c.
- Forgetting that after day T the series starts from S_T, not from 0: the term $(1+k)^m S_T$ is easy to drop.
- Saying the debt is paid off because "d is bigger than c". What matters is d against the interest kS_T at opening time.

Beyond STEP

The recurrence $S_n = aS_{n-1} + b$ is the simplest *linear first-order difference equation*. Its solution is a fixed point $b/(1-a)$ plus a multiple of a^n, which is exactly the shape of our formula: d/k plus a multiple of $(1+k)^m$. Whether the multiple is positive or negative decides between growing debt and repayment, an idea that reappears in the stability of fixed points and in linear ODEs. Boyce and DiPrima, *Elementary Differential Equations*, treats difference equations alongside differential ones, and Siklos, *Advanced Problems in Mathematics*, has more STEP practice on recurrences.

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